

AIM Programs for Causal Inference

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April 10, 2026

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1 Terminology

This preliminary section introduces the minimal terminology for statistics and causality required to define, contextualize, and analyze AIM programs.

1.1 Statistical Modelling

1.2 Causal Modelling

1.3 Causal Programs

2 AIM Programs

A particular class of functions that can inhabit the type signature $\Theta \times \mathcal{U} \rightarrow \mathcal{D}$ are AIM programs.

2.1 Grammar

AIM programs consist of sequences of the following three terms, each of which has type signature $(\Theta \times \mathcal{U}) \rightarrow (\Theta \times \mathcal{D})$

2.2 Semantics

here's what AIM programs due
note commutativity of same operators but not diff ones in general
show why each heterogenous pari doesnt commute?

2.3 Comparative Expressivity

compare the exptrssivity of AIm programs to other programs

3 AIM Programs Over Graphical Models

Here's how AIM programs get additional structure via the do calculus axioms when you have a graph model

I-elim rewrites
heterogenous commutativity

4 Causal Identifiability via AIM Programs

in general identifiability means that models that agree about all purely probabilistic models agree for a causal query - or equivalently that the causal query is a function of probabilistic queries.

pearl says that if a causal queries expression as a probabilistic query cannot be reduced to elim the i operator it is not identifiable (clearly if it can be reduced it is identifiable)

analogous result for rewrites directly in AIM?

5 General Inference via AIM Programs

Even if an AIM program's causal component is not reducible (identifiable), we can still perform interesting analysis

5.1 Evaluation of Statistical Estimators

show the markov chain, talk about performance definitions of estimators on it (bayesian and non bayesian)

5.2 Synthesis of Statistical Estimators

talk about ML estimates, mention broader estimates

5.3 Information Theoretic Measures

entropy, reduction in entropy, and mutual information, and channel capacity

6 Related Work

talk about related work

7 Future Work

7.1 Mechanized Proofs

embed AIM semantics in a setting amenable to mechanized proofs

7.2 AIM Language Expansion

extend language to have more general branching and constructs

7.3 AIM Program Expressivity/Contextualization

compare to other languages, determine non-transferrable embeddings

7.4 Concretize Modelling and Semantics

pick Θ , talk about particular cases of inference wrt it